

Competitive Programming Midterm Progress Report

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June 21, 2026

1 Introduction

This report summarizes my progress during the first four weeks of Competitive Programming practice. The objective of this phase was to build a strong foundation in problem solving, algorithmic thinking, and efficient implementation using C++.

The training included structured practice sheets, CSES problem sets, Codeforces practice, and topic-wise learning resources. Through consistent problem solving, I improved my understanding of fundamental competitive programming concepts and developed the ability to approach problems systematically.

2 Week 1: Foundations of Competitive Programming

2.1 Topics Covered

- C++ STL Basics
- Time Complexity Analysis
- Prefix Sums
- Difference Arrays
- Mathematical Implementation Problems
- Conditional Logic and Basic Problem Solving

2.2 Key Learnings

- Learned efficient usage of STL containers and algorithms.
- Understood the importance of time complexity and asymptotic analysis.

- Learned how prefix sums can optimize range query problems.
- Learned the concept of difference arrays for efficient range updates.
- Improved implementation and debugging skills.

2.3 Problems Solved

During Week 1, I successfully completed both practice sheets provided in the course.

- Sheet 1: 26 problems
- Sheet 2: 14 problems

A total of **40 problems** were solved during this week. These problems covered implementation, mathematics, conditionals, sorting, and introductory algorithmic techniques.

3 Week 2: Prefix Sums, Frequency Counting and Greedy Thinking

3.1 Topics Covered

- Prefix Sum Applications
- Frequency Counting using Hash Maps
- Two Sum Variants
- Sorting Techniques
- Greedy Problem Solving
- Introduction to Codeforces Div 2 A/B Problems

3.2 Key Learnings

- Solving subarray problems using prefix sums.
- Using frequency maps for efficient counting.
- Understanding greedy observations and interval scheduling.
- Developing logical thinking required for Codeforces Div 2 problems.
- Improving implementation under contest-style constraints.

3.3 Problems Solved

Solved several classical CSES and practice problems, including:

- Distinct Numbers
- Apartments
- Ferris Wheel
- Concert Tickets
- Restaurant Customers
- Movie Festival
- Sum of Two Values
- Subarray Sums I
- Subarray Sums II
- Subarray Divisibility

These problems strengthened my understanding of sorting, greedy methods, prefix sums, and frequency-based techniques.

4 Week 3: Greedy Algorithms and Binary Search

4.1 Topics Covered

- Greedy Algorithms
- Exchange Argument Technique
- Binary Search
- Binary Search on Answer
- Searching and Sorting Techniques

4.2 Key Learnings

- Understanding why greedy solutions work.
- Identifying optimal greedy choices.
- Applying binary search to answer spaces instead of arrays.
- Recognizing monotonic properties in optimization problems.
- Improving problem decomposition skills.

4.3 Practice

- CSES Searching and Sorting Problems
- Codeforces Greedy Problems
- Codeforces Constructive Problems

5 Week 4: Bit Manipulation and Number Theory

5.1 Topics Covered

- Bit Manipulation
- Binary Representation
- XOR Properties
- Prime Numbers
- Prime Factorization
- Modular Arithmetic
- Introductory Number Theory

5.2 Key Learnings

- Efficient use of bitwise operators.
- Solving problems using XOR observations.
- Understanding prime factorization techniques.
- Applying modular arithmetic to large computations.
- Using mathematical observations to simplify solutions.

5.3 Practice

Practice was focused on Codeforces, LeetCode, AtCoder, and SPOJ problems related to bit manipulation and number theory.

6 Challenges Faced

- Initially finding the correct observation for greedy problems.
- Learning how to optimize solutions to avoid Time Limit Exceeded errors.
- Understanding when to use prefix sums versus hash-map based approaches.
- Improving speed during contests while maintaining correctness.
- Becoming comfortable with unfamiliar problem statements.

7 Progress and Achievements

One of the most significant outcomes of these four weeks has been measurable improvement in competitive programming performance.

- Completed both Week 1 practice sheets (40 problems).
- Solved multiple CSES Searching and Sorting problems.
- Practiced Codeforces Div 2 A/B level problems.
- Participated in several contests held on Codeforces and OAHelper.
- Strengthened understanding of Greedy Algorithms, Binary Search, Prefix Sums, Bit Manipulation, and Number Theory.
- Improved my rating upto 903 on Codeforces.

8 Key Takeaways

- Competitive programming requires both algorithmic knowledge and strong observation skills.
- Many problems are solved through identifying patterns rather than applying complex algorithms.
- Consistent practice is more important than solving only difficult problems.
- Careful analysis of mistakes is essential for long-term improvement.
- Contest experience helps improve speed, confidence, and decision making.

9 GitHub Repository

All solution codes and practice work have been maintained in a GitHub repository.

GitHub Repository:

<https://github.com/parthingle007-dev/Competitive-Programming-SOC/tree/main>

Codeforces Profile:

<https://codeforces.com/profile/pi7>

10 Future Goals

- Reach a four-digit Codeforces rating.
- Become comfortable solving most Div 2 A and B problems independently.
- Improve proficiency in Greedy Algorithms and Binary Search.
- Begin studying Graph Algorithms and Dynamic Programming.
- Participate regularly in contests and improve contest performance.

11 Conclusion

The first four weeks provided a strong foundation in competitive programming. Through structured learning resources, regular practice, and contest participation, I significantly improved my problem-solving skills and competitive programming performance.

The completion of both Week 1 sheets, successful practice on CSES and Codeforces, participation in contests, and an increase in Codeforces rating from 0 to 903 demonstrate consistent effort and measurable growth. These experiences have strengthened my analytical thinking and problem-solving abilities, and I look forward to building on these fundamentals in the coming weeks.

Overall the resources provided are really very helpful by the mentors and I have been enjoying my time learning and testing myself in the program.